

PORTFOLIO CASE STUDY

Bitcoin vs. Bitcoin Proxy Stocks

An AI-Augmented Investment Analytics System

A June 2025 analysis rebuilt with a ChatGPT-centered five-stage decision process

Central question: Should an investor own Bitcoin directly, or can a publicly traded Bitcoin proxy create greater value per diluted common share during a major Bitcoin bull market?

Scope

Six public companies | Three Bitcoin scenarios | 24-month horizon | Source data through August 2026

Prepared as an employer-facing demonstration of decision framing, source normalization, financial modeling, validation, and executive interpretation.

Technical summary

Decision: A proxy opportunity exists. MSTR is the best overall choice when the goal is the highest credible probability of beating Bitcoin; ABTC has the greatest modeled upside; Metaplanet offers the strongest middle ground.

The analysis evaluates six Bitcoin-linked public companies against the benchmark of simply owning BTC. The primary KPI is modeled stock return minus Bitcoin return under identical price scenarios. The model is per-share and capital-structure aware: total BTC holdings alone are not accepted as evidence of shareholder value creation.

Three companies beat Bitcoin in all three modeled scenarios: MSTR, Metaplanet, and American Bitcoin. Twenty One Capital, CleanSpark, and MARA produce positive stock returns in some cases, but fail the actual benchmark because they underperform BTC after dilution, senior claims, capex, mining difficulty, and execution requirements are incorporated.

Rank	Proxy	Role	BTC +100% stock return	Excess vs BTC	Scenarios beating BTC
1	MSTR	Best overall	128.4%	+28.4 pts	3 / 3
2	Metaplanet	Best risk-adjusted alternative	124.9%	+24.9 pts	3 / 3
3	ABTC	Maximum convexity	246.5%	+146.5 pts	3 / 3
4	XXI	Discount special situation	93.9%	-6.1 pts	0 / 3
5	CLSK	Diversified miner	32.9%	-67.1 pts	0 / 3
6	MARA	High-beta trading vehicle	56.8%	-43.2 pts	0 / 3

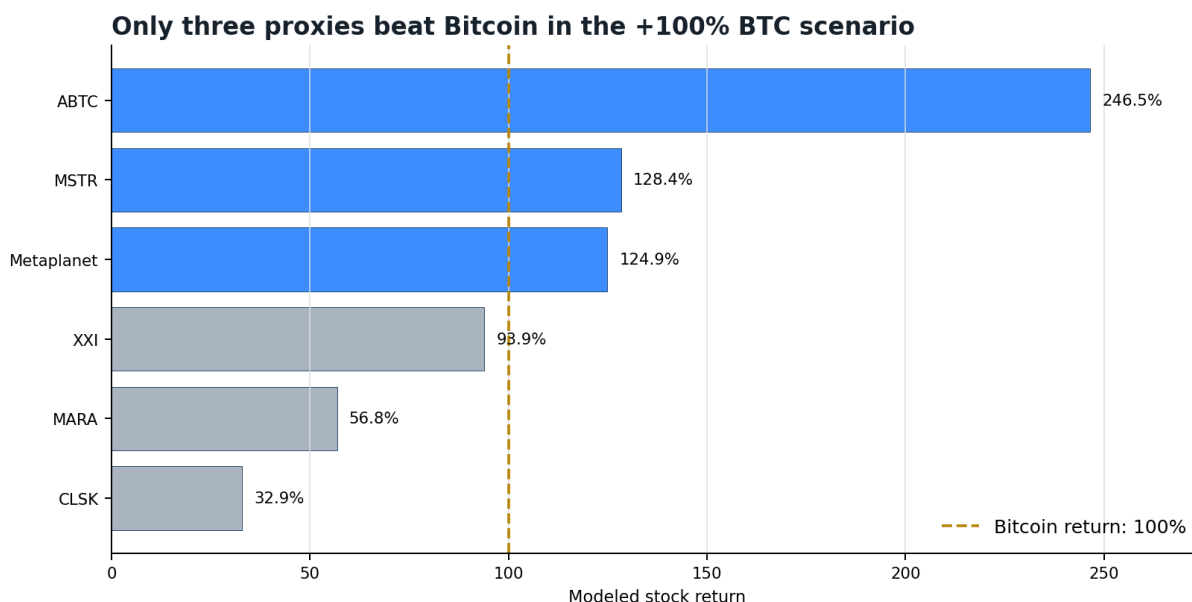


Figure 1. Modeled stock returns when Bitcoin rises 100%.

Source: Stage 4 rebuilt model; valuation anchor BTC \$78,511.23; 24-month horizon.

Why this project was rebuilt with AI

The project originated as a more code-heavy analysis completed around June 2025. This version deliberately tests whether the same investment question can be handled through a ChatGPT-centered workflow without abandoning analytical discipline. The AI does not replace the investment decision. It accelerates the structure, research synthesis, normalization, modeling, validation, and communication surrounding that decision.

Earlier project	AI-augmented redevelopment
Code-led exploration	Decision-led five-stage workflow
Company screening and historical analysis	Explicit benchmark, hypotheses, KPI definitions, and source hierarchy
Manual analytical assembly	ChatGPT-assisted source normalization and auditable scenario model
Results emphasized	Results, assumptions, model checks, caveats, and decision implications preserved together
Static project output	Workbook, notebook, detailed report, and presentation as linked portfolio artifacts

Employer takeaway: The project demonstrates how AI can extend an analyst's judgment and execution capacity while keeping assumptions visible, calculations reproducible, and recommendations bounded by evidence.

The five-stage analytical system

Stage	Purpose	Output
1. Start	Identify the real decision alternatives	Own BTC; own a proxy; reject the proxy opportunity
2. Framing	Convert the alternatives into one decision question	Can a proxy create more value per diluted share than BTC itself?
3. Design	Define evidence before collecting it	Hypotheses, KPIs, segments, confounders, and required metrics
4. Execution	Build the source-backed model	Normalized inputs, 18 scenario rows, checks, and reproducible notebook
5. Finish	Interpret the evidence and recommend	Ranking, decision, uncertainty, and monitoring rules

Stage 1 - Start: define the decision before analyzing companies

The business problem is not 'which company owns the most Bitcoin?' It is which exposure best expresses a bullish Bitcoin thesis after accounting for company-specific risks.

Decision A - Own Bitcoin directly

Receive essentially pure BTC exposure without dilution, management, operating, financing, or capital-allocation risk. This is the controlling benchmark, not a passive comparison point.

Decision B - Own a publicly traded Bitcoin proxy

Accept additional risk in exchange for potential outperformance created by operating leverage, fixed senior claims, BTC-per-share accretion, NAV discount closure, or accretive financing.

Decision C - Reject the proxy opportunity

Conclude that no stock offers enough expected excess return to compensate for dilution, debt, operating costs, capex, execution risk, or governance risk.

Stage 1 decision: Should the investor own Bitcoin directly, or is there a public proxy with a sufficiently attractive and defensible economic mechanism to justify choosing its common stock instead?

Stage 2 - Framing: force every proxy to beat the benchmark

The unit of analysis is the economic outcome for one diluted common share. Total BTC, Bitcoin-related revenue, historical beta, or headline treasury growth can all rise while value per share deteriorates.

Primary analytical question: Among publicly traded businesses materially exposed to Bitcoin, are there companies whose economics can outperform BTC during a major bull market after accounting for dilution, liabilities, operating costs, capital requirements, and execution risk?

Every candidate must answer one question: Why should an investor accept this company's additional risks instead of simply buying Bitcoin?

Stage 3 - Design: determine what evidence would change the decision

Hypotheses

- H1: At least one proxy has an economically defensible mechanism capable of creating greater value per diluted share than direct BTC ownership.
- H0: After dilution, senior claims, capex, mining difficulty, financing needs, and execution risk, proxies do not offer a sufficiently reliable advantage.
- Treasury hypothesis: net BTC per diluted share matters more than total BTC growth.
- Mining hypothesis: efficient miners outperform only when price-driven margin expansion exceeds difficulty, energy, capex, and dilution growth.
- NAV hypothesis: discount closure can create convexity, but it is less reliable than demonstrated per-share value creation.
- Capital-allocation hypothesis: high BTC correlation can coexist with poor shareholder economics.

Primary and secondary decision metrics

Metric	Definition	Decision use
Modeled excess return vs BTC	Proxy stock return minus BTC return under the same scenario	Directly tests whether company risk is rewarded
Probability of beating BTC	Judgment band based on dilution, execution, capital structure, operations, and valuation	Separates maximum upside from likelihood of realization
BTC per diluted share	BTC holdings divided by diluted common shares	Tests whether treasury growth reaches each shareholder
Net BTC per diluted share	BTC and liquid assets less senior claims, divided by diluted shares	Captures capital-structure burden

Evidence architecture and source normalization

The evidence layer combines market data, company filings, investor-relations disclosures, Bitcoin network data, and explicit model assumptions. Source-backed inputs remain distinct from forecast assumptions, and missing fields are carried as model risks rather than silently set to zero.

Source family	Evidence captured	Role in the model	Main limitation
Token Terminal	BTC price, market context, fee mix	Benchmark anchor and miner revenue context	Not a source for company capital structures
Mempool / network research	Difficulty, hashrate, revenue mix	Mining competition assumptions	Snapshot rather than full forecast
Company IR and SEC filings	BTC, shares, debt, preferreds, ATM, mining and HPC	Company-specific model inputs	Disclosure timing differs by issuer
Uploaded prior analysis	Earlier targets, risk hypotheses, probability judgments	Context and challenge set	Contained denominator conflicts and subjective ranges
Rebuilt workbook + notebook	Normalized metrics, formulas, checks, scenarios	Controlling analytical source	Forecasts remain assumption-driven

The source-refresh workbook records the evidence class, as-of date, URL, limitation, and workbook use for each input family. Two denominator conflicts identified in the earlier analysis were corrected before Stage 4 was completed.

Stage 4 - Execution: rebuild the model around company-specific economics

The execution layer produces 18 company-scenario observations: six companies multiplied by three BTC outcomes. It deliberately avoids a single generic valuation formula.

Component	Treasury companies	Bitcoin miners
Core value	Terminal BTC assets plus liquid assets	Terminal treasury value plus mining enterprise value
Claims	Debt, preferreds, convertibles, and fixed charges	Debt, leases, equipment obligations, and financing
Dilution	Equity issuance and price-dependent conversion	ATM utilization and stock compensation
Operating value	Normally separated from BTC reserve value	Production, difficulty, cash cost, capex, and EBITDA

Component	Treasury companies	Bitcoin miners
Optionality	NAV discount or premium behavior	HPC value kept separate and probability-weighted
Per-share output	Common equity value divided by ending diluted shares	Treasury + mining + HPC less net claims, divided by ending shares

Scenario assumptions

Scenario	Terminal BTC	BTC return	Difficulty increase	Horizon
BTC +50%	\$117,766.85	+50%	+25%	24 months
BTC +100%	\$157,022.46	+100%	+45%	24 months
BTC +200%	\$235,533.69	+200%	+80%	24 months

Conservative design choice: Miner profitability does not scale one-for-one with Bitcoin. Rising price attracts hashrate and capital, so difficulty, cash cost, capex, and dilution are increased alongside BTC.

Model validation and execution gate

The workbook contains a visible Checks sheet and a separate calculation notebook. All twelve execution assertions pass: company count, scenario count, model-row count, populated targets, positive common equity, positive ending shares, representative return ties, sats/share ties, equity bridges, notebook calculations, and the Stage 4 completion gate.

Validation item	Result
Model observations	18 of 18 rebuilt
Missing target prices	0
Formula errors	0 detected
Independent notebook checks	6 of 6 passed
Execution status	STAGE 4 COMPLETE

Stage 4 results: only half the proxy universe beats Bitcoin

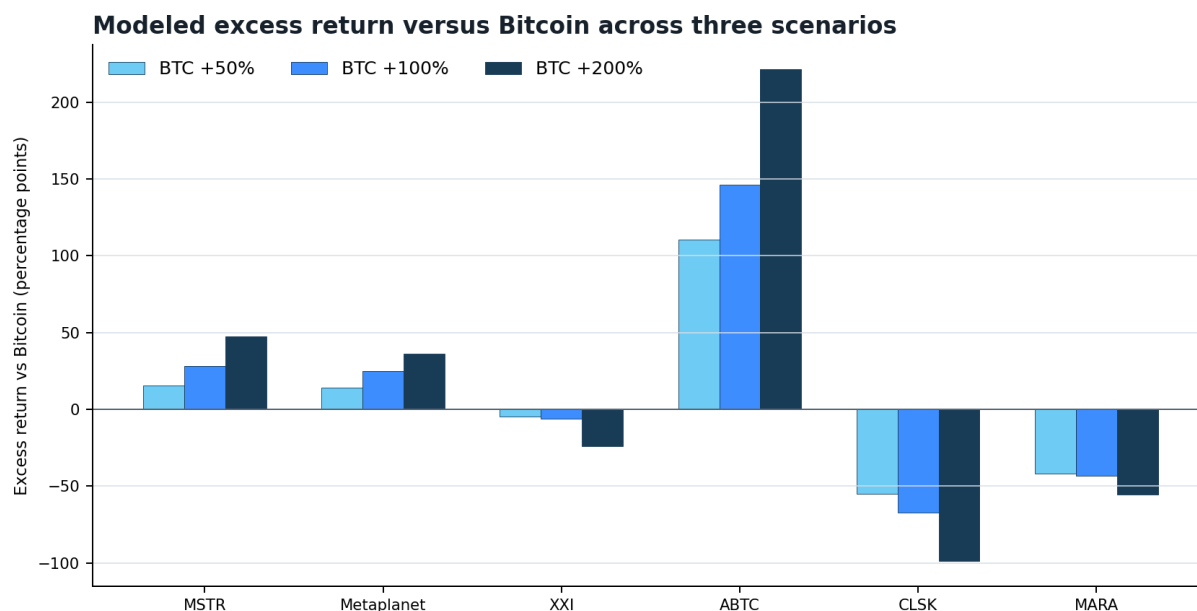


Figure 2. Modeled stock return minus Bitcoin return across all three BTC scenarios.

Source: Stage 4 rebuilt scenario model. Positive values indicate that accepting company risk is rewarded in the modeled case.

Company	Target +50%	Excess +50%	Target +100%	Excess +100%	Target +200%	Excess +200%
MSTR	\$209.72	+15.4 pts	\$289.65	+28.4 pts	\$440.69	+47.5 pts
Metaplanet*	\$3.40	+14.2 pts	\$4.66	+24.9 pts	\$6.97	+36.4 pts
XXI	\$9.47	-4.5 pts	\$12.63	-6.1 pts	\$17.96	-24.1 pts
ABTC	\$24.09	+110.7 pts	\$32.02	+146.5 pts	\$48.20	+221.6 pts
CLSK	\$12.09	-54.9 pts	\$16.89	-67.1 pts	\$25.54	-99.0 pts
MARA	\$12.81	-41.7 pts	\$18.55	-43.2 pts	\$28.90	-55.7 pts

*Metaplanet targets are shown in USD equivalent using the model's same-date FX conversion. Local-market presentation should use the corresponding JPY values after refreshing FX.

Per-share exposure explains why headline BTC growth is insufficient

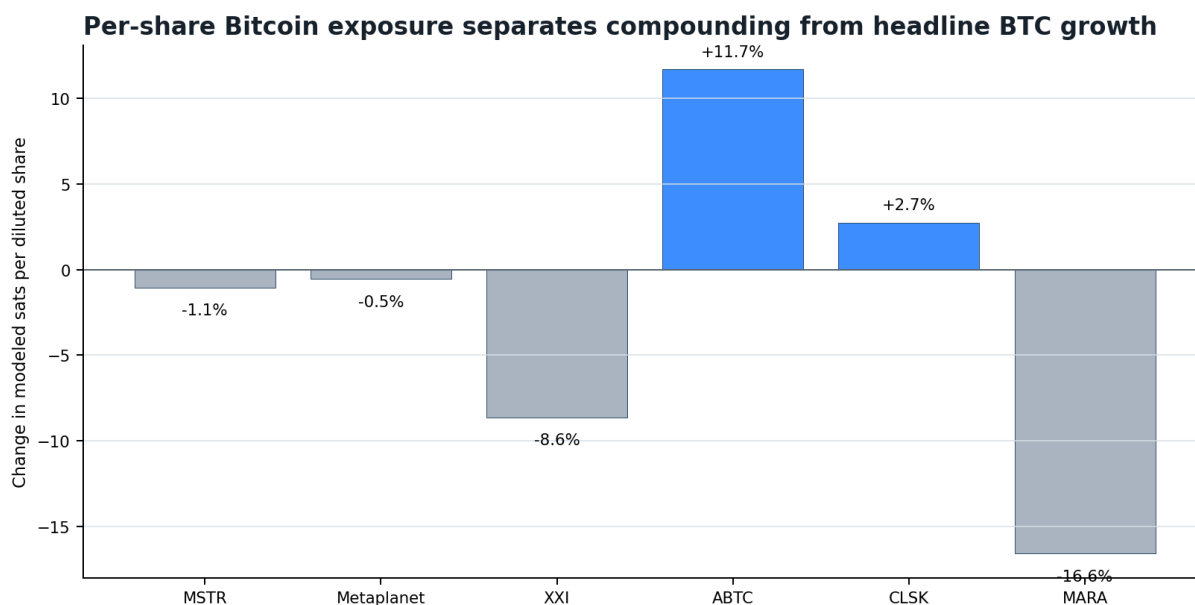


Figure 3. Change in modeled sats per diluted share in the BTC +100% scenario.

Source: Stage 4 rebuilt model and normalized starting denominators.

ABTC is the only candidate combining large modeled stock outperformance with substantial growth in BTC per diluted share. MSTR and Metaplanet still outperform because fixed-claim leverage and financing mechanics offset modest dilution. CLSK produces slight sats/share growth but fails to translate it into superior equity returns after capital intensity. MARA's falling BTC-per-share profile is the weakest transmission mechanism in the group.

Company interpretation

1. MSTR - highest-confidence leveraged Bitcoin

MSTR beats Bitcoin in all three scenarios. Its excess return rises from 15.4 points in the +50% case to 47.5 points in the +200% case. The mechanism is straightforward: a large BTC asset base appreciates while debt, preferred claims, and other fixed obligations do not rise proportionally. Strategy also has the most proven capital-markets execution platform in the group.

What could break the thesis: Continued issuance that reduces BTC per share, expensive preferred financing, cash held rather than deployed, or a lower NAV multiple.

2. Metaplanet - strongest valuation and quality asymmetry

Metaplanet also beats Bitcoin in all three modeled cases, with slightly lower excess return than MSTR. The attraction is the combination of BTC exposure, fixed-claim leverage, and a more favorable starting valuation. It becomes the leading risk-adjusted alternative if BTC-per-share growth continues and its full capital stack confirms the implied NAV bridge.

What could break the thesis: An incomplete cash/liability bridge, warrant or preferred dilution, derivative losses, FX changes, or persistent NAV discount.

3. American Bitcoin - maximum fundamental convexity

ABTC is the numerical winner. In the +100% BTC scenario the stock rises 246.5%, beating Bitcoin by 146.5 percentage points. It combines treasury appreciation, mining-margin expansion, and a relatively small starting equity denominator. Modeled sats per share rise 11.7% in that scenario.

What could break the thesis: The \$2.1 billion ATM program, higher difficulty, equipment and lease obligations, pledged BTC, fleet obsolescence, or costs above model assumptions.

4. Twenty One Capital - discounted Bitcoin without proven compounding

XXI remains a potentially attractive NAV-discount special situation, but the rebuilt model reverses the earlier outperformance thesis. Once conversion and share-count mechanics are incorporated, it underperforms Bitcoin in all three scenarios. Discount closure is a market rerating assumption, not demonstrated operating value creation.

5. CleanSpark - a stronger company than Bitcoin proxy

CLSK may be the strongest diversified operator among the miners, but that is not the same as being the strongest BTC proxy. Difficulty, capex, debt, dilution, and HPC funding absorb too much of the modeled BTC upside. HPC can add independent value, while simultaneously reducing the purity of the Bitcoin exposure.

6. MARA - high beta with the weakest per-share transmission

MARA can remain a powerful trading vehicle, but the fundamental model does not show reliable BTC outperformance. BTC sales, debt, ATM issuance, infrastructure spending, and a 16.6% modeled decline in sats per share in the +100% scenario weaken the link between Bitcoin's success and common-share value.

Stage 5 - Finish: interpret the evidence and make the decision

Final decision: Choose Decision B selectively. Use MSTR as the primary proxy when the goal is the highest credible probability of outperforming Bitcoin. Use ABTC only when maximum upside is more important than dilution and execution risk. Metaplanet is the strongest middle ground.

Rank	Company	Investment role	Estimated probability of beating BTC*	Decision
1	MSTR	Highest-confidence leveraged BTC	60-70%	Primary proxy
2	Metaplanet	Best valuation/quality asymmetry	55-65%	Risk-adjusted alternative
3	ABTC	Maximum convexity	50-60%	Aggressive allocation only
4	XXI	Discount special situation	35-45%	Watchlist
5	CLSK	Diversified mining/HPC	30-40%	Good company, weak benchmark fit
6	MARA	High-beta BTC trade	25-35%	Reject for fundamental proxy thesis

*Probability bands are structured forward-looking judgments, not statistical estimates. Historical adjusted-price data were insufficient for a defensible empirical probability model.

What would change the ranking

- Move Metaplanet ahead of MSTR if its capital stack is fully reconciled and BTC per diluted share continues growing while MSTR's declines.
- Move ABTC toward the top if it uses the ATM selectively, sustains BTC-per-share accretion, and outgrows network difficulty without cost deterioration.
- Upgrade XXI only after a price-dependent diluted cap table is confirmed and management demonstrates sustained BTC-per-share growth.
- Upgrade miners only when hashrate growth, efficiency, power economics, and financing jointly support free-cash-flow accretion per share.
- Default back to direct BTC if proxy valuation, capital allocation, or disclosure quality makes the excess-return mechanism too dependent on unverified assumptions.

Limitations, uncertainty, and robustness

The analysis is ready to share with caveats. The calculations reconcile, but the forecasts are not company guidance and do not establish a statistical probability distribution. The conclusions are conditional on the explicit 24-month BTC, difficulty, financing, dilution, cost, and valuation assumptions.

Issue	Status	Decision impact
Metaplanet full cash and liability bridge	Partial	Could change NAV and ranking
Miner 24-month operating guidance	Modeled assumptions	Could materially change production, cost, and EBITDA
Historical return distributions	Partial	Probability bands remain subjective
HPC contract timing and funding	Modeled separately	Could change CLSK/MARA value but not proxy purity
Taxes and jurisdiction	Not fully modeled	May affect after-tax comparability

Issue	Status	Decision impact
Rapidly changing prices and capital structures	Refresh required	Targets and ranking can move materially

Portfolio capabilities demonstrated

Capability	Demonstrated work
Decision framing	Converted a vague 'best Bitcoin proxy' request into three alternatives and one benchmarked analytical question
Metric design	Defined excess return vs BTC as the primary KPI and per-share capital economics as the controlling unit
Research synthesis	Combined market, network, filings, IR, capital structure, mining, and HPC evidence
Data normalization	Corrected denominator conflicts and distinguished basic, diluted, and economic share counts
Financial modeling	Built company-specific treasury, mining, conversion, dilution, and sum-of-parts mechanics
Reproducibility	Delivered a formula-driven workbook and executed calculation notebook with explicit checks
Validation	Ran formula, bridge, denominator, target-price, and notebook tie-outs before interpretation
Executive communication	Separated highest upside from highest probability and translated evidence into a clear decision
AI augmentation	Used ChatGPT to orchestrate the five-stage process while preserving human judgment and auditability

Reproducibility and project artifacts

The project is designed as a connected portfolio package rather than a single presentation. Each artifact has a distinct analytical role:

- Bitcoin_Proxy_Stage5_Input.xlsx - normalized evidence, model assumptions, 18 scenario rows, summary outputs, gaps, sources, and checks.
- Stage4_Model_Rebuild.ipynb - reproducible companion calculations and independent validation checks.
- AI_Augmented_Bitcoin_Proxy_Analysis_Report.pdf - detailed employer-facing case study.
- AI_Augmented_Bitcoin_Proxy_Analysis_Presentation.pptx - concise interview and LinkedIn/GitHub presentation.

Recommended repository structure: README overview -> report and deck -> workbook and notebook -> data/source notes. The AI-augmented redevelopment should appear in the README and both deliverables, not only in a comment.

Source and audit trail

Bitcoin market metrics (2026-08-25) - Token Terminal: <https://tokenterminal.com/terminal/projects/bitcoin/metrics>

Bitcoin network data (2026-08-14 to 2026-08-21) - Mempool: <https://mempool.space/>

Strategy filings and investor materials (2026-08-23) - Strategy IR: <https://www.strategy.com/investor-relations>

Metaplanet filings and shareholder materials (2026 Q2/H1) - Metaplanet IR: <https://metaplanet.jp/en/shareholders>

American Bitcoin filings (2026 Q2) - American Bitcoin IR: <https://ir.americanbitcoin.com/financials/sec-filings>

Twenty One Capital filings (2026 Q1/latest) - SEC EDGAR:

<https://www.sec.gov/edgar/search/#/q=Twenty%2520One%2520Capital>

CleanSpark filings (2026 Q2/July) - CleanSpark IR: <https://investors.cleanspark.com/financials/sec-filings/default.aspx>

MARA filings (2026 Q2) - MARA IR: <https://ir.mara.com/financial-information/sec-filings>

Model output source: Bitcoin_Proxy_Stage5_Input.xlsx and Stage4_Model_Rebuild.ipynb. Prior contextual analysis: Pasted markdown(20260827-031908).md. Company disclosures change quickly; exact filing accession and page references should be refreshed before publication as investment research.

Conclusion

The project demonstrates that AI-augmented analytics is not merely faster reporting. Used correctly, it is a disciplined decision workflow: define alternatives, frame the benchmark, design the evidence, execute an auditable model, and interpret the result without confusing theoretical upside with probability of realization.

Final answer: The proxy opportunity should not be rejected. MSTR is the best overall expression of the thesis, Metaplanet is the strongest risk-adjusted alternative, and ABTC offers the greatest upside if its dilution and operating risks remain controlled.